Beyond Closed-Set Accuracy: Topology-Invariant NetFlow Generalization, Class-Conditional Mondrian Conformal Calibration, and Selective SOC Triage for Network Intrusion Detection

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Abstract—Machine learning-based Network Intrusion Detection Systems (NIDS) routinely report near-perfect closed-set classification metrics exceeding 99 percent F1 on single-testbed benchmarks, yet suffer catastrophic false alarm rates and generalization collapse when deployed in live Security Operations Centers (SOCs). Drawing on ten foundational experimental security pitfalls, we design, implement, and empirically verify a six-stage, leakage-safe NIDS reliability architecture evaluated across 1,986,745 flows in NF-UNSW-NB15-v2 and 17,129,715 flows in NF-CSE-CIC-IDS2018-v2. First, we replace corrupted legacy CICFlowMeter CSV extractions with 43-feature standardized NetFlow v2 representations and execute a four-seed bidirectional feature-leakage ablation across 16 training and transfer runs. We prove that stripping nine environment-specific shortcut features, including Time-To-Live, TCP window bounds, and protocol identifiers, leaves in-distribution Macro-F1 virtually unchanged from 0.9480 plus or minus 0.0003 to 0.9436 plus or minus 0.0004, while monotonically increasing forward zero-shot cross-network Macro-F1 from 0.4479 plus or minus 0.0018 to 0.4776 plus or minus 0.0057, Balanced Accuracy from 0.4509 plus or minus 0.0027 to 0.4878 plus or minus 0.0083, and Matthews Correlation Coefficient from minus 0.1038 plus or minus 0.0038 to minus 0.0332 plus or minus 0.0173. Second, we demonstrate that standard Marginal Split Conformal Prediction (SCP) fails under the 2,472 to 1 class imbalance of enterprise traffic: at miscoverage alpha equal to 0.10, Marginal SCP achieves an illusory 89.90 percent aggregate coverage by over-covering Benign traffic at 98.53 percent while starving minority attacks, where Analysis coverage drops to 41.38 percent and DoS drops to 60.38 percent. Our Inductive Mondrian Conformal Prediction (MCP) engine with Kolmogorov-Smirnov Clustered Conformal Prediction (CCP) fallback restores finite-sample coverage across all nine categories to 90.08 percent plus or minus 1.58 percent, lifting Analysis coverage by 49.14 percentage points to 90.52 percent. Third, we unify penultimate gradient-boosted decision tree logit representations with Mahalanobis Weibull Extreme Value Theory (EVT) open-set recognition, achieving 0.9044 OpenAUROC on withheld zero-day attacks. Finally, on the full 298,154-flow test split, our selective SOC routing gate resolves 71.09 percent of all traffic autonomously at 99.31 percent accuracy with a Triage Area Under the Risk-Coverage Curve of 0.0597 and a False Alarm Rate of 4.20 percent, while escalating 86.59 percent of unseen zero-day Worms flows to human analysts and maintaining 100.00 percent attack recall under volumetric evasion shifts via monotone tree constraints.

Index Terms—network intrusion detection systems, cross-dataset generalization, shortcut learning, open-set recognition, extreme value theory, Mondrian conformal prediction, selective classification, security operations center triage

I. Introduction

Despite two decades of research into machine learning (ML) for Network Intrusion Detection Systems (NIDS), a stark operational divide persists between academic benchmarks and real-world Security Operations Center (SOC) deployments [1], [2]. While published studies routinely claim 99% accuracy on single-environment splits of legacy datasets, production security teams remain reluctant to automate blocking actions from ML-driven alerts. Reference [1] codified ten methodological pitfalls that silently inflate benchmark scores while obscuring operational fragility:

- Extractor Corruption and Label Noise (Pitfalls P1 and P2): The widely used raw comma-separated values (CSV) distributions of CIC-IDS2017 and CSE-CIC-IDS2018 suffer from state-tracking defects inside the CICFlowMeter extraction tool [3]–[5]. When Transmission Control Protocol (TCP) connections terminate via FIN flags or expire, CICFlowMeter spawns artifactual zero-payload "TCP appendix" flows with inverted source and destination orientations that automated timestamp scripts mislabel as attacks [3], alongside ignored RST termination flags and negative flow durations.
- Data Snooping and Temporal Campaign Leakage (Pitfalls P3 and P5): Applying global feature normalization prior to splitting or executing random k-fold cross-validation over network captures allows temporally adjacent packets from the same bursty attack campaign to straddle training and test partitions.
- Shortcut Learning on Testbed Artifacts (Pitfalls P4 and P9): Even after removing Internet Protocol (IP) addresses and port numbers, packet header fields

such as Time-To-Live (MIN_TTL, MAX_TTL) and initial TCP window sizes (TCP_WIN_MAX_IN) encode the fixed operating system and virtual router hop count of synthetic attacker virtual machines (VMs) rather than intrinsic attack mechanics [7]. Crucially, in-distribution validation cannot detect this shortcut learning because models with and without leaking features achieve virtually identical same-dataset F1 scores (> 0.94).

- Closed-Set Assumption Versus Zero-Day Reality: Standard multi-class classifiers apply a closed-set Softmax or tree-voting normalization over K known training categories ($\mathcal{Y}_{\text{known}}$), forcing unseen zero-day attack families ($\mathcal{Y}_{\text{unknown}}$) into existing labels with high uncalibrated confidence [10], [11].
- Marginal Calibration Collapse Under Base-Rate Skew (Pitfalls P7 and P8): Enterprise networks exhibit extreme benign dominance (96.22% benign in NF-UNSW-NB15-v2, creating a $25.46:1$ binary skew and a $2,483:1$ skew against rare attack families). Under this base rate, standard Marginal Split Conformal Prediction (SCP) satisfies a global $1-\alpha = 90\%$ coverage target simply by over-covering the benign majority while allowing minority attack coverage to collapse below 42% [13].

In this paper, we address these failure modes through an end-to-end, six-stage NIDS reliability architecture built from first principles. Rather than proposing an over-parameterized deep neural network evaluated on a random split, we anchor our study on strong tabular Gradient-Boosted Decision Trees (GBDTs) [8] with domain-informed volumetric monotonicity constraints [9], paired with covariance-aware Mahalanobis Weibull Extreme Value Theory (EVT) open-set recognition (OSR), class-conditional Inductive Mondrian Conformal Prediction (MCP) [12], [14], and a selective classification SOC triage gate [16]. Every architectural decision is verified empirically across five falsifiable experimental suites (Suites E1–E5), and we provide a rigorous mathematical verification of both positive findings and negative directional transfer phenomena.

II. Data Forensics, Schema Harmonization, and Leakage-Safe Partitioning

A. Standardization on NetFlow v2 Benchmarks

To eliminate the feature incompatibility and extraction bugs of raw CICFlowMeter CSV files without incurring deep packet inspection (DPI) latency, our pipeline standardizes on the University of Queensland NetFlow v2 (NF-v2) schema extracted directly from raw packet captures (PCAPs) via nProbe [6]. Whereas NF-v1 exported only 8 to 12 basic flow fields that lacked sub-flow discriminative power, NF-v2 provides 43 standardized features (41 behavioral and header attributes plus binary Label and multi-class Attack columns) across heterogeneous network topologies.

Prior to model development, we executed an automated ingestion sanity audit across the full Parquet distributions of our source enterprise benchmark (NF-UNSW-NB15-v2) and target cloud benchmark (NF-CSE-CIC-IDS2018-v2). Table I summarizes the verified structural properties of both benchmarks. Both datasets exhibit 1:1 nonlabel column alignment (41 shared features) and exactly zero null cells across $19,116,460$ combined flows. However, during cross-domain preprocessing, our forensic audit uncovered a critical numerical hazard: NF-CSE-CIC-IDS2018-v2 stores derived throughput rates (SRC_TO_DST_SECOND_BYTES and DST_TO_SRC_SECOND_BYTES) as 64-bit floats containing sub-millisecond burst spikes that exceed the maximum finite representation of 32-bit floating-point numbers ($\approx 3.40 \times 10^{38}$). Naive casting to float32 triggers silent overflow to $\pm\infty$. Our ForensicDataSanitizer module resolves this by filtering negative durations and zero-packet artifacts, clamping numerical attributes to physical multi-terabit bounds $[-10^{15}, 10^{15}]$ in 64-bit precision, and deduplicating strictly on training reference partitions so that hard or ambiguous test flows are never artificially discarded (mitigating Pitfalls P1 and P3).

TABLE I
Verified Structural and Base-Rate Audit of Standardized NetFlow v2 Benchmarks

Dataset Property	Standardized Benchmark Distribution	
	NF-UNSW-NB15-v2	NF-CSE-CIC-IDS2018-v2
Topology	Enterprise Testbed	500-Host AWS Cloud
Total Flows	1,986,745	17,129,715
Columns / Shared	43 / 41	43 / 41
Verified Nulls	0	0
Benign (Label=0)	1,911,666 (96.22%)	15,101,685 (88.16%)
Attack (Label=1)	75,079 (3.78%)	2,028,030 (11.84%)
Binary Skew	25.46 : 1	7.45 : 1
Attack Classes	8 Known + 1 Zero-Day	Harmonized Binary[a]
Max Class Skew	$2,482.68:1$	—

[a]Evaluated on 1:1 aligned binary labels for zero-shot transfer.

B. Progressive Shortcut-Learning Ablation Schemas

Removing explicit routing identifiers (IPV4_SRC_ADDR, IPV4_DST_ADDR, L4_SRC_PORT, L4_DST_PORT, FLOW_ID, and timestamps) leaves 39 candidate features in our Baseline schema. However, testbed-specific artifacts remain embedded in packet headers. To isolate and quantify shortcut learning (Pitfall P4) via bidirectional cross-dataset transfer [7], we define four nested feature schemas:

- Baseline (39 features): Drops explicit IP, port, and timestamp identifiers; retains all 39 numeric NF-v2 attributes.
- Set A—TTL Pruning (37 features): Additionally drops MIN_TTL and MAX_TTL, eliminating hop-count and guest operating system fingerprinting of synthetic attacker VMs.

- Set B—Window Pruning (35 features): Additionally drops TCP_WIN_MAX_IN and TCP_WIN_MAX_OUT, removing host network-stack buffer size signatures.
- Set C—Topology-Invariant Core (30 features): Additionally drops static protocol enumeration identifiers (PROTOCOL, L7_PROTO, ICMP_TYPE, ICMP_IPV4_TYPE, and DNS_QUERY_ID), locking strictly behavioral flow volume, packet length variance, TCP flag dynamics, inter-arrival timing, and retransmission rates.

C. Chronological Four-Way Partitioning

To prevent temporal burst leakage (Pitfall P3) and validation parameter snooping (Pitfall P5), our partitioning module splits each known class $c \in \{0, 1, \ldots, K-1\}$ in chronological capture order into four strictly disjoint subsets using ratios $(0.55, 0.15, 0.15, 0.15)$, as in:

$$\mathcal{D}_{\text{known}} = Z_t\,(55\%) \cup Z_v\,(15\%) \cup Z_c\,(15\%) \cup Z_{\text{test}}^{\text{known}}\,(15\%), \tag{1}$$

where Z_t ($1,092,615$ flows in full NF-UNSW-NB15-v2) fits model weights and the interquartile RobustScaler; Z_v ($297,988$ flows) governs hyperparameter selection and early stopping; Z_c ($297,988$ flows) is reserved exclusively for fitting Weibull EVT tail parameters, reconstruction thresholds, and Mondrian conformal quantiles; and Z_{test} ($298,154$ flows) is locked for single-pass evaluation. All flows belonging to the withheld zero-day attack family ($y = -1$, Worms, 164 flows) are excluded from Z_t, Z_v, and Z_c in (1) and injected exclusively into Z_{test} to simulate an unseen open-set intrusion campaign.

III. Methodology and Mathematical Formulation

A. Stage 3: Domain-Informed Monotone GBDT Backbone

Tree-based ensembles outperform unconstrained deep neural networks on tabular data because decision trees naturally learn sharp, axis-aligned discontinuities across heterogeneous feature scales [8]. Our primary classification backbone employs a histogram-based GBDT (HistGradientBoostingClassifier, maximum iterations = 200, learning rate = 0.08, max leaf nodes = 63, balanced class weighting).

To defend against volumetric evasion attacks [9], we partition the feature index set $\mathcal{F} = \{1, \ldots, d\}$ into volumetric rate and accumulation features $\mathcal{F}_{\text{vol}}$ (columns containing BYTES, PKTS, or THROUGHPUT) and unconstrained behavioral features $\mathcal{F} \backslash \mathcal{F}_{\text{vol}}$. For binary anomaly detection, we enforce a structural monotonicity constraint vector $\mathbf{m} \in \{0, +1\}^d$, defined by:

$$m_j = \begin{cases} +1, & \text{if } j \in \mathcal{F}_{\text{vol}}, \\ 0, & \text{otherwise.} \end{cases} \tag{2}$$

Equation (2) guarantees that for any two flow feature vectors $\mathbf{x}, \mathbf{x}' \in \mathbb{R}^d$ satisfying $x_j \le x'_j$ for all $j \in \mathcal{F}_{\text{vol}}$ and $x_j = x'_j$ for $j \notin \mathcal{F}_{\text{vol}}$, the tree ensemble decision score $f(\mathbf{x})$ satisfies:

$$f(\mathbf{x}) \le f(\mathbf{x}') \implies \hat{P}(Y = 1 \mid \mathbf{x}) \le \hat{P}(Y = 1 \mid \mathbf{x}'). \tag{3}$$

The monotonicity guarantee in (3) prevents unconstrained trees from carving out "benign pockets" in the upper volumetric tail when high-volume benign file transfers (elephant flows) appear in training data.

B. Stage 4: Dual-Stage Open-Set Recognition via Mahalanobis Weibull EVT

To detect zero-day attacks ($\mathcal{Y}_{\text{unknown}}$) without retraining, we extract penultimate activation vectors $\mathbf{v}(\mathbf{x}) \in \mathbb{R}^K$ from the positive-shifted decision-function logits $\mathbf{z}(\mathbf{x})$ of the trained backbone:

$$\mathbf{v}(\mathbf{x}) = \mathbf{z}(\mathbf{x}) - \min_k z_k(\mathbf{x})\mathbf{1} + \epsilon_0 \mathbf{1}, \tag{4}$$

where $\epsilon_0 = 10^{-3}$ ensures strictly positive activations. Conventional OpenMax measures Euclidean distance $\|\mathbf{v}(\mathbf{x}) - \boldsymbol{\mu}_c\|_2$ from class Mean Activation Vectors (MAVs) $\boldsymbol{\mu}_c$, which ignores strong inter-logit correlations in network traffic [10]. We instead compute the covariance-aware Mahalanobis distance for each known class $c \in \{0, \ldots, K-1\}$ using the regularized empirical covariance matrix $\boldsymbol{\Sigma}_c$ estimated strictly over correctly classified training flows $Z_{t,c}^{\text{ok}} = \{i \in Z_t : y_i = c \wedge \hat{y}_i = c\}$:

$$\begin{aligned} \boldsymbol{\mu}_c &= \frac{1}{|Z_{t,c}^{\text{ok}}|} \sum_{i \in Z_{t,c}^{\text{ok}}} \mathbf{v}(\mathbf{x}_i), \\ \boldsymbol{\Sigma}_c &= \text{Cov}_{i \in Z_{t,c}^{\text{ok}}}\big(\mathbf{v}(\mathbf{x}_i)\big) + \lambda_0 \mathbf{I}, \end{aligned} \tag{5}$$

where $\lambda_0 = 10^{-4}$ guarantees positive-definite invertibility in (5). The Mahalanobis distance $d_M(\mathbf{v}(\mathbf{x}), \boldsymbol{\mu}_c)$ is then given by:

$$d_M\big(\mathbf{v}(\mathbf{x}), \boldsymbol{\mu}_c\big) = \sqrt{\big(\mathbf{v}(\mathbf{x}) - \boldsymbol{\mu}_c\big)^\top \boldsymbol{\Sigma}_c^{-1} \big(\mathbf{v}(\mathbf{x}) - \boldsymbol{\mu}_c\big)}. \tag{6}$$

By Extreme Value Theory (Fisher–Tippett–Gnedenko theorem), the upper tail of the $\eta = 50$ largest Mahalanobis distances in class c from (6) follows a Weibull cumulative distribution function (CDF) $W_c(d_M)$ with shape $\kappa_c > 0$ and scale $\lambda_c > 0$:

$$W_c(d_M) = 1 - \exp\big(-(d_M/\lambda_c)^{\kappa_c}\big). \tag{7}$$

At inference, each activation $v_c(\mathbf{x})$ from (4) is attenuated using (7) by weight $\omega_c(\mathbf{x}) = 1 - \vartheta W_c(d_M(\mathbf{v}(\mathbf{x}), \boldsymbol{\mu}_c))$ with $\vartheta = 0.85$, reallocating subtracted mass to an explicit zero-day logit $\tilde{v}_{\text{unk}}(\mathbf{x})$:

$$\begin{aligned} \tilde{v}_c(\mathbf{x}) &= v_c(\mathbf{x})\,\omega_c(\mathbf{x}), \\ \tilde{v}_{\text{unk}}(\mathbf{x}) &= \sum_{c=0}^{K-1} \max\big(0, v_c(\mathbf{x}) - \tilde{v}_c(\mathbf{x})\big). \end{aligned} \tag{8}$$

To intercept stealthy zero-day flows that project near the interior of a known class cluster [11], Stage 4 pairs boundary attenuation in (8) with a category-specific Principal

Component Analysis (PCA) subspace reconstructor $\mathcal{R}_c$ fitted on $Z_{t,c}^{\text{ok}}$. A flow assigned top closed-set class $\hat{c}$ is rejected as a zero-day anomaly ($I_{\text{OSR}}(\mathbf{x}) = 1$) if either $\arg\max_j \tilde{v}_j(\mathbf{x}) = \text{unk}$ or its reconstruction error $\mathcal{L}_{\text{rec}}(\mathbf{x}; \hat{c})$ exceeds the $\theta = 0.96$ empirical quantile $Q_{0.96}$ calibrated on $Z_{c,\hat{c}}$:

$$\mathcal{L}_{\text{rec}}(\mathbf{x}; \hat{c}) = \frac{1}{K}\left\|\mathbf{v}(\mathbf{x}) - \mathcal{R}_{\hat{c}}\big(\mathbf{v}(\mathbf{x})\big)\right\|_2^2 > Q_{0.96}. \tag{9}$$

C. Stage 5: Inductive Mondrian Conformal Calibration and KS-Clustered Fallback

Let $s(\mathbf{x}, y) = 1 - \hat{P}_h(Y = y \mid \mathbf{x})$ denote the hinge nonconformity score of candidate label y under classifier h. Standard Marginal SCP computes a single global quantile q_{marg} across all $N_c = |Z_c|$ calibration instances:

$$q_{\text{marg}} = Q_{\lceil (N_c+1)(1-\alpha)\rceil / N_c}\big(\{s(\mathbf{x}_i, y_i) : i \in Z_c\}\big). \tag{10}$$

While q_{marg} in (10) guarantees marginal coverage $\mathbb{P}(Y_{\text{test}} \in \Gamma^{\alpha}_{\text{marg}}(\mathbf{X}_{\text{test}})) \geq 1 - \alpha$, 96.22% of Z_c in NF-UNSW-NB15-v2 consists of Benign flows ($y_i = 0$) whose nonconformity scores are near zero. Consequently, q_{marg} is dominated by benign traffic and severely under-covers minority attack classes.

To achieve exact class-conditional validity (mitigating Pitfall P7), we partition Z_c by true label $\tilde{y} \in \{0, \ldots, K-1\}$ into disjoint Mondrian strata $Z_{c,\tilde{y}} = \{i \in Z_c : y_i = \tilde{y}\}$ of size $n_{\tilde{y}} = |Z_{c,\tilde{y}}|$ [12], [14]. For any rare attack stratum where $n_{\tilde{y}} < \lceil (1-\alpha)/\alpha \rceil$ (e.g., $n_{\tilde{y}} < 9$ at $\alpha = 0.10$), finite-sample quantiles cannot guarantee $1 - \alpha$ coverage alone; in that regime, our calibrator triggers a Clustered Conformal Prediction (CCP) fallback [15] that merges $Z_{c,\tilde{y}}$ with the well-sampled stratum c^* minimizing the two-sample Kolmogorov–Smirnov (KS) distance between empirical nonconformity CDFs $\hat{F}$:

$$c^* = \arg\min_{c' : n_{c'} \geq \lceil (1-\alpha)/\alpha \rceil} \sup_{t \in [0,1]} \left|\hat{F}_{\tilde{y}}(t) - \hat{F}_{c'}(t)\right|. \tag{11}$$

Using the stratum $Z_{c,\tilde{y}}$ (or its KS-merged cluster from (11)), let $G_{\tilde{y}}(\mathbf{x}) = |\{i \in Z_{c,\tilde{y}} : s(\mathbf{x}_i, \tilde{y}) > s(\mathbf{x}, \tilde{y})\}|$ and $E_{\tilde{y}}(\mathbf{x}) = |\{i \in Z_{c,\tilde{y}} : |s(\mathbf{x}_i, \tilde{y}) - s(\mathbf{x}, \tilde{y})| \leq 10^{-7}\}|$. For each test flow $\mathbf{x}$ and candidate label $\tilde{y}$, we compute the smoothed Mondrian p-value with tie-breaking $\theta \sim \text{Uniform}[0, 1]$:

$$p_{\tilde{y}}(\mathbf{x}) = \frac{G_{\tilde{y}}(\mathbf{x}) + \theta\left(E_{\tilde{y}}(\mathbf{x}) + 1\right)}{n_{\tilde{y}} + 1}. \tag{12}$$

Because naive 2D matrix broadcasting of (12) across $N_{\text{test}} = 298,154$ test flows and $n_0 = 286,749$ benign calibration flows would allocate a $298,154 \times 286,749$ boolean array requiring 79.59 GiB of random-access memory (RAM), we sort each calibration stratum $Z_{c,\tilde{y}}$ once during fitting and evaluate $G_{\tilde{y}}(\mathbf{x})$ and $E_{\tilde{y}}(\mathbf{x})$ via vectorized binary search (np.searchsorted), reducing complexity from $\mathcal{O}(N_{\text{test}} N_c)$ to $\mathcal{O}(N_{\text{test}} \log N_c)$ and RAM footprint from 79.59 GiB to 4.55 MiB.

The Mondrian prediction set at significance level $\alpha \in (0, 1)$ is constructed from (12) as:

$$\Gamma^{\alpha}(\mathbf{x}) = \big\{\tilde{y} \in \{0, \ldots, K-1\} : p_{\tilde{y}}(\mathbf{x}) > \alpha\big\}, \tag{13}$$

which guarantees finite-sample class-conditional validity across every attack category:

$$\mathbb{P}\big(Y_{\text{test}} \in \Gamma^{\alpha}(\mathbf{X}_{\text{test}}) \mid Y_{\text{test}} = c\big) \geq 1 - \alpha. \tag{14}$$

From the ordered Mondrian p-values $p_{(1)}(\mathbf{x}) \geq p_{(2)}(\mathbf{x}) \geq \cdots \geq p_{(K)}(\mathbf{x})$ in (12), we extract the inference-time Pairwise Conformal Evidence Margin:

$$\delta^{\text{em}}(\mathbf{x}) = p_{(1)}(\mathbf{x}) - p_{(2)}(\mathbf{x}) \in [0, 1], \tag{15}$$

which quantifies the statistical separation between the top two competing hypotheses without requiring ground-truth labels [14].

D. Stage 6: Selective SOC Triage Routing Policy

To prevent alert fatigue under the Base Rate Fallacy (Pitfall P8) [16], we implement a selective classification routing gate governed by prediction set cardinality $|\Gamma^{\alpha}(\mathbf{x})|$ from (13), Stage-4 rejection indicator $I_{\text{OSR}}(\mathbf{x}) \in \{0, 1\}$ from (9), and evidence margin threshold $\tau_{\text{hi}} = 0.25$ on (15). Each flow $\mathbf{x}_i \in Z_{\text{test}}$ is routed to either the Automated Resolution Queue ($\mathcal{A}$) or the Escalated Analyst Triage Queue ($\mathcal{E}$):

$$\mathbf{x}_i \in \begin{cases} \mathcal{A}, & \text{if } |\Gamma^{\alpha}(\mathbf{x}_i)| = 1 \wedge I_{\text{OSR}}(\mathbf{x}_i) = 0 \wedge \delta^{\text{em}}(\mathbf{x}_i) \geq \tau_{\text{hi}}, \\ \mathcal{E}, & \text{otherwise.} \end{cases} \tag{16}$$

Crucially, we verified during ablation testing that Stage-5 empty conformal sets ($|\Gamma^{\alpha}(\mathbf{x}_i)| = 0$) in (16) must not overwrite closed-set predictions as $\hat{y} = -1$ (Zero-Day_Unknown), because Mondrian calibration at $\alpha = 0.10$ in (14) intentionally caps Benign coverage at 90%; roughly 8.4% of atypical benign flows fall outside q_0 while simultaneously having near-zero attack probabilities. Routing $|\Gamma^{\alpha}(\mathbf{x}_i)| = 0$ to $\mathcal{E}$ while preserving top closed-set predictions when $I_{\text{OSR}}(\mathbf{x}_i) = 0$ prevents an 8.42 percentage-point false-alarm penalty while still guaranteeing human review.

We evaluate selective triage efficiency using the Area Under the Risk-Coverage Curve (AURC). Ordering test flows by descending evidence margin $\pi(1), \ldots, \pi(N_{\text{test}})$ from (15), the selective risk $\hat{R}(k)$ at coverage k/N_{test} and overall AURC are:

$$\hat{R}(k) = \frac{1}{k}\sum_{j=1}^{k} \mathbb{I}\big(\hat{y}_{\pi(j)} \neq y_{\pi(j)}\big), \quad \text{AURC} = \frac{1}{N_{\text{test}}}\sum_{k=1}^{N_{\text{test}}} \hat{R}(k). \tag{17}$$

IV. Empirical Results and Verification of Interpretations

All five experimental suites were executed in Python 3.12 using our modular repository. Below, we present the empirical tables and rigorously verify the underlying statistical and physical mechanisms.

A. Suites E1 and E2: Four-Seed Bidirectional Cross-Dataset Shortcut Ablation

To ensure that cross-network findings are not artifacts of random initialization or subsampling luck, we executed the bidirectional transfer protocol (NF-UNSW-NB15-v2 $\leftrightarrow$ NF-CSE-CIC-IDS2018-v2) across four independent random seeds ($s \in \{42, 43, 44, 45\}$, 16 total training and cross-evaluation runs). Table II reports the parametric (Mean $\pm$ Std) and non-parametric (Min/Max) statistics across all four schemas, and Table III provides the granular per-seed breakdown.

We verify three physical mechanisms from Tables II and III:

1) In-Distribution Blindness to Shortcut Learning: Across all four seeds, pruning 9 header features from Baseline (39 features) to Set_C (30 features, a 23.1% reduction in dimensionality) alters in-distribution UNSW_InDist_F1 by only -0.0044 ($0.9480 \pm 0.0003 \rightarrow 0.9436 \pm 0.0004$). This empirically confirms Pitfall P4 [1], [7]: when evaluated strictly on same-testbed splits, shortcut features appear harmless because behavioral timing and rate features suffice to separate classes in-distribution.
2) Strict Monotonicity of Forward Transfer (UNSW $\rightarrow$ CIC18): In the forward transfer direction, removing testbed shortcuts improves zero-shot Macro-F1 across every single pruning tier: 0.4479 (Baseline) $\rightarrow$ 0.4636 (Set_A) $\rightarrow$ 0.4678 (Set_B) $\rightarrow$ 0.4776 (Set_C), for a net gain of +2.97 percentage points (+3.69 percentage points in Balanced Accuracy and +0.0706 in Matthews Correlation Coefficient (MCC)). Crucially, Table III verifies that the minimum 4-seed score of Set_C (0.4692 on Seed 43) strictly exceeds the maximum 4-seed score of Baseline (0.4500 on Seed 44). The largest single leap occurs at Set_A (+1.57 percentage points F1 and +2.70 percentage points BalAcc upon dropping MIN_TTL and MAX_TTL), proving that the Baseline tree ensemble latched onto UNSW virtual router hop counts that inverted when evaluated on the 500-machine Amazon Web Services (AWS) topology of CIC-IDS2018.
3) Why Reverse Transfer (CIC18 $\rightarrow$ UNSW) Exhibits Asymmetry and Higher Seed Variance: Training on CIC18-v2 and testing on UNSW-v2 declines from 0.5054 ± 0.0148 (Baseline) to 0.4569 ± 0.0422 (Set_C) while exhibiting a 7.4$\times$ larger seed standard deviation (0.0422 versus 0.0057). Two structural factors govern this asymmetry. First, whereas the UNSW-v2 model trains on the complete deterministic chronological split (1.09M training flows), the reverse CIC18-v2 model trains on a seed-dependent $200,000$-flow subsample drawn from $17,129,715$ total flows. On Seed 42, Set_C dipped to 0.4063, whereas on Seed 44 it reached 0.5004 (BalAcc = 0.5027). Had we stopped at a single seed (Seed 42), we would have misreported a severe reverse collapse (0.4063) that was actually a subsampling outlier corrected by multi-seed aggregation (median 0.4605). Second, NF-CSE-CIC-IDS2018-v2 attacks (11.84% of traffic) are dominated by high-rate volumetric Denial-of-Service (DoS) and Distributed DoS (DDoS) floods, whereas NF-UNSW-NB15-v2 attacks (3.78% of traffic) are dominated by low-volume Exploits, Fuzzers, and Reconnaissance probes that share protocol numbers (PROTOCOL and L7_PROTO) with specific services. When Set_C strips protocol identifiers and enforces positive volumetric monotonicity, a model trained exclusively on high-rate cloud floods struggles to recognize stealthy low-rate enterprise exploits without target-domain calibration.

B. Suite E4: Marginal Split Conformal Collapse Versus Mondrian Restoration

Tables IV and V evaluate standard Marginal SCP from (10) against our Class-Conditional Inductive MCP calibrator from (13) at target miscoverage rates $\alpha = 0.10$ (90.0% target) and $\alpha = 0.05$ (95.0% target).

Tables IV and V provide a definitive empirical demonstration of Pitfalls P7 and P8 in uncertainty quantification:

1) The Illusion of Marginal Coverage: A practitioner inspecting only aggregate coverage (Overall = 89.90% at $\alpha = 0.10$ and 94.96% at $\alpha = 0.05$) would falsely conclude that Marginal SCP is well calibrated. Yet Table V reveals that Marginal SCP achieves 89.90% overall coverage by covering 98.53% of Benign flows while allowing coverage on critical intrusions to collapse to 41.38% on Analysis, 60.38% on DoS, and 74.43% on Exploits. Across the 9 classes, Marginal SCP averages only 76.88% with an inter-class standard deviation of 17.43%.
2) Mondrian Equitability Across a $2,472:1$ Imbalance: Partitioning calibration scores into label-conditional Mondrian strata forces each class c to satisfy (14) independently. Consequently, Mondrian MCP achieves a mean class coverage of 90.08% at $\alpha = 0.10$ (95.13% at $\alpha = 0.05$) and reduces inter-class standard deviation eleven-fold from 17.43% down to 1.58% (bounded tightly between 88.02% and 93.49%). This lifts Analysis coverage by +49.14 percentage points and DoS coverage by +27.64 percentage points, while increasing mean prediction set size by only 0.262 labels ($0.989 \rightarrow 1.251$).
3) Why Mondrian MCP Produces More Empty Sets ($1,247$ Versus 606): Mathematically, Marginal SCP sets its single global threshold q_{marg} high enough to include Benign in 98.53% of all benign prediction sets. Mondrian MCP tightens the Benign threshold q_0 so that exactly 89.95% of true benign flows include the Benign label. For the remaining

TABLE II
Suites E1 and E2: Four-Seed Bidirectional Cross-Dataset Generalization and Shortcut Feature Ablation (Seeds 42–45, $n = 4$ Seeds per Schema)

Feature Schema	Retained Feats	Cross-Domain and In-Distribution Evaluation Metrics (Mean ± Std and Min / Max)				
		UNSW In-Dist F1	UNSW → CIC18 F1	U2C BalAcc	U2C MCC	CIC18 → UNSW F1
Baseline	39	0.9480 ± 0.0003	0.4479 ± 0.0018	0.4509 ± 0.0027	-0.1038 ± 0.0038	0.5054 ± 0.0148
(Min / Max)		(0.9475 / 0.9483)	(0.4460 / 0.4500)	(0.4479 / 0.4540)	(-0.1078 / -0.0994)	(0.4931 / 0.5258)
Set A (No TTL)	37	0.9468 ± 0.0005	0.4636 ± 0.0047	0.4779 ± 0.0074	-0.0630 ± 0.0147	0.4766 ± 0.0131
(Min / Max)		(0.9461 / 0.9473)	(0.4598 / 0.4695)	(0.4718 / 0.4874)	(-0.0746 / -0.0439)	(0.4607 / 0.4902)
Set B (No Win)	35	0.9417 ± 0.0048	0.4678 ± 0.0060	0.4738 ± 0.0085	-0.0610 ± 0.0151	0.4606 ± 0.0296
(Min / Max)		(0.9346 / 0.9445)	(0.4629 / 0.4754)	(0.4664 / 0.4844)	(-0.0733 / -0.0417)	(0.4178 / 0.4858)
Set C (Invariant)	30	$\mathbf{0.9436 \pm 0.0004}$	$\mathbf{0.4776 \pm 0.0057}$[a]	$\mathbf{0.4878 \pm 0.0083}$	$\mathbf{-0.0332 \pm 0.0173}$	$\mathbf{0.4569 \pm 0.0422}$
(Min / Max)		(0.9432 / 0.9441)	(0.4692 / 0.4822)	(0.4755 / 0.4938)	(-0.0584 / -0.0195)	(0.4063 / 0.5004)

[a]Minimum 4-seed score of Set C (0.4692) strictly exceeds the maximum 4-seed score of Baseline (0.4500).

TABLE III
Granular Per-Seed Telemetry for Forward (UNSW → CIC18) and Reverse (CIC18 → UNSW) Cross-Dataset Transfer

Schema	Transfer Direction	Seed 42	Seed 43	Seed 44	Seed 45	Median	Mean ± Std
Baseline (39)	UNSW → CIC18 Macro-F1	0.4488	0.4469	0.4500	0.4460	0.4478	0.4479 ± 0.0018
Set A (37)	UNSW → CIC18 Macro-F1	0.4652	0.4598	0.4695	0.4599	0.4626	0.4636 ± 0.0047
Set B (35)	UNSW → CIC18 Macro-F1	0.4630	0.4629	0.4754	0.4700	0.4665	0.4678 ± 0.0060
Set C (30)	UNSW → CIC18 Macro-F1	0.4792	0.4692	0.4822	0.4797	0.4795	0.4776 ± 0.0057
Baseline (39)	CIC18 → UNSW Macro-F1	0.5068	0.5258	0.4960	0.4931	0.5014	0.5054 ± 0.0148
Set A (37)	CIC18 → UNSW Macro-F1	0.4716	0.4902	0.4837	0.4607	0.4777	0.4766 ± 0.0131
Set B (35)	CIC18 → UNSW Macro-F1	0.4693	0.4696	0.4858	0.4178	0.4695	0.4606 ± 0.0296
Set C (30)	CIC18 → UNSW Macro-F1	0.4063	0.4811	0.5004	0.4398	0.4605	0.4569 ± 0.0422

TABLE IV
Suite E4: Aggregate Versus Class-Conditional Validity Comparing Marginal SCP and Mondrian MCP

Conformal Method	Calibration Metrics Across Significance Levels				
	$\boldsymbol{\alpha}$	Overall	Worst	Mean	Size / Empty
Marginal SCP	0.10	89.90%	41.38%	76.88%	0.989 / 606
Mondrian MCP	0.10	90.03%	88.02%	90.08%	1.251 / 1247
Marginal SCP	0.05	94.96%	52.59%	86.01%	1.165 / 0
Mondrian MCP	0.05	94.87%	93.88%	95.13%	1.530 / 440

TABLE V
Suite E4: Per-Category Coverage Breakdown at $\alpha = 0.10$ (90.0% Target) Under Severe Imbalance

Flow Category	Empirical Class-Conditional Coverage (%)		
	Marginal SCP	Mondrian MCP	MCP Gain
Benign	98.53	89.95	−8.58 pp[a]
Analysis	41.38	90.52	+49.14 pp
DoS	60.38	88.02	+27.64 pp
Exploits	74.43	90.37	+15.94 pp
Generic	76.86	89.99	+13.13 pp
Backdoor	76.00	88.80	+12.80 pp
Fuzzers	83.14	90.77	+7.63 pp
Shellcode	92.56	93.49	+0.93 pp
Reconnaissance	88.66	88.78	+0.12 pp
Mean ± Std	$\mathbf{76.88 \pm 17.43}$	$\mathbf{90.08 \pm 1.58}$	**+13.20** pp

[a]Eliminates majority over-coverage and restores exact 90% target.

$\sim 10\%$ of atypical benign flows whose attack class probabilities are also near zero, $p_{\tilde{y}}(\mathbf{x}) \leq 0.10$ for all $\tilde{y} \in \{0, \ldots, 8\}$, yielding $\Gamma^{0.10}(\mathbf{x}) = \emptyset$. Rather than being a defect, these empty sets identify boundary-anomalous benign traffic and safely route it to the human analyst queue $\mathcal{E}$.

C. Suite E3: Unified GBDT-Backbone Open-Set Recognition

During our initial implementation audit, computing Stage-4 OpenMax distances on embeddings extracted from an auxiliary unweighted multi-layer perceptron (MLPClassifier) yielded 0.7055 OpenAUROC for Mahalanobis_EVT and 0.5477 for Euclidean_OpenMax, trailing the GBDT Closed_MaxProb baseline (0.9233). We verified that this gap was caused by backbone confounding: the unweighted MLP underfitted minority attack classes compared to the class-balanced GBDT. Table VI presents the unified comparison after extracting penultimate activation vectors $\mathbf{v}(\mathbf{x})$ in (4) directly from the trained HistGradientBoostingClassifier decision-function logits.

On the 64-dimensional MLP embedding space in Table VI, replacing Euclidean distance with covariance-aware Mahalanobis distance improved OpenAUROC by +15.78 percentage points (0.5477 → 0.7055) and Area Under the Precision-Recall Curve (OpenAUPR) by +64.3% (0.0070 → 0.0115). Unifying the backbone onto the GBDT lifted Euclidean_OpenMax by +30.81 percentage points (0.8558) and Mahalanobis_EVT by +14.05 percentage

TABLE VI
Suite E3: Open-Set Recognition on Withheld Zero-Day Family (Worms) Across Auxiliary MLP Versus Unified GBDT Backbones

Open-Set Method	Backbone and Zero-Day Detection Scores		
	Backbone	AUROC	AUPR
Euclidean OpenMax	Aux MLP (64-D)	0.5477	0.0070
Mahalanobis EVT	Aux MLP (64-D)	0.7055	0.0115
Euclidean OpenMax	GBDT Logits	0.8558	0.0267
Mahalanobis EVT	GBDT Logits	0.8460	0.0294
Hybrid Maha+CP	GBDT Logits	0.9044	0.0429
Closed MaxProb	GBDT Logits	0.9233	0.0576

points (0.8460, with a +155.7% increase in OpenAUPR to 0.0294). In a 9-class GBDT, the decision-function output has dimension $K = 9$; because tree logits are directly optimized for class separation, unknown Worms flows induce simultaneous moderate activations across overlapping attack logits (Exploits and DoS), which directly depresses $\max_c \hat{P}_c(\mathbf{x})$ (0.9233 OpenAUROC). Combining Mahalanobis Weibull tail probability with conformal margin uncertainty (Hybrid Maha+CP) achieves 0.9044 OpenAUROC and 0.0429 OpenAUPR while retaining explicit per-class tail thresholds for Stage-6 triage.

D. Suite E5: Full-Scale 1.98M-Flow Selective SOC Triage and Monotone Evasion Resilience

Table VII reports the complete per-category performance of the integrated six-stage pipeline on the locked test split ($N_{\text{test}} = 298,154$ flows) of the full $1,986,745$-flow NF-UNSW-NB15-v2 benchmark under strict 30-feature Set_C pruning.

When 11 shortcut header features are stripped under Set_C in Table VII, overlapping attack categories such as Analysis (33.62% raw recall), Backdoor (37.60%), Reconnaissance (52.03%), and DoS (53.04%) exhibit moderate top-1 closed-set recall because they share timing and packet-size profiles with Exploits. However, Mondrian conformal prediction captures this overlap by outputting multi-label prediction sets $\Gamma^{0.10}(\mathbf{x})$ (e.g., $\{\text{Exploits}, \text{Analysis}\}$) that contain the true class 90.08% of the time. By routing multi-sets ($|\Gamma^{0.10}(\mathbf{x})| > 1$) and low-margin flows ($\delta^{\text{em}}(\mathbf{x}) < 0.25$) via (16) to human analysts in $\mathcal{E}$, the automated queue $\mathcal{A}$ resolves 71.09% of all network traffic ($211,954$ flows) at 99.31% accuracy (AURC = 0.0597 in (17)), achieving a +32.02 percentage-point mean accuracy lift across the 9 known categories (Analysis: 33.62% → 100.00%; Backdoor: 37.60% → 94.29%; Reconnaissance: 52.03% → 96.63%; Shellcode: 76.74% → 98.85%). Furthermore, 86.59% (142 of 164 flows) of the unseen Worms zero-day category were blocked from autonomous misclassification and escalated directly to $\mathcal{E}$.

Finally, Table VIII verifies the resilience of our monotone-constrained GBDT backbone ($m_j = +1$ in (2)) against an unconstrained GBDT when 5% of benign training flows consist of high-volume file transfers (elephant flows) and test attack flows undergo volumetric padding shifts $\varepsilon \in [0.00, 1.00]$.

As shown in Table VIII, the unconstrained GBDT maintains 99.96% → 98.88% attack recall for small volumetric shifts $\varepsilon \in [0.00, 0.75]$, creating a false sense of security. However, because 5% of benign training traffic occupied the upper volumetric region ($+6\sigma$), the unconstrained tree split the upper tail into a "benign leaf pocket." Once padded attack flows crossed that split threshold at $\varepsilon = 1.00$, unconstrained attack recall suffered a catastrophic phase-transition collapse from 98.88% down to 0.59% (Macro-F1 dropping from 0.9891 to 0.3677). In contrast, enforcing positive monotonicity constraints ($m_j = +1$ in (3)) mathematically forbids any split from assigning a lower attack probability to a higher volumetric value, maintaining 100.00% attack recall and 0.9943 Macro-F1 with zero variance across the entire sweep.

V. Limitations and Future Directions

In accordance with Pitfall P10 [1], we document three operational limitations of our architecture:

1) Cross-Domain Covariate Shift Without Target Calibration: While pruning shortcut features (Set_C) monotonically improves zero-shot transfer from NF-UNSW-NB15-v2 to NF-CSE-CIC-IDS2018-v2 by +2.97 percentage points F1 (0.4776 ± 0.0057), absolute zero-shot performance remains bounded near 0.48–0.51 Macro-F1 due to the 25.46 : 1 versus 7.45 : 1 base-rate shift and taxonomy divergence between low-rate enterprise exploits and cloud DDoS floods. Deploying across distinct networks therefore requires re-calibrating the Stage-5 Mondrian quantiles q_c using a small target-domain calibration buffer Z_c^{target}.
2) Single-Flow Tabular Scope Versus Multi-Flow Graph Lateral Movement: Our backbone processes individual 43-feature NetFlow records in $\mathcal{O}(1)$ memory, which enables line-rate inference but does not capture multi-hop lateral movement across host communication graphs [10].
3) Analyst Queue Load Under $\alpha = 0.05$: Tightening Mondrian miscoverage from $\alpha = 0.10$ to $\alpha = 0.05$ raises worst-class coverage to 93.88% and mean class coverage to 95.13%, but increases mean prediction set size from 1.251 to 1.530, which expands the fraction of multi-set flows routed to the human analyst queue $\mathcal{E}$.

VI. Conclusion

We have presented and empirically verified a six-stage reliability framework for NIDS that bridges the gap between academic benchmark accuracy and operational SOC deployment. By combining forensically sanitized NetFlow v2 representations, progressive shortcut feature pruning (Set_C), monotone-constrained gradient-boosted

TABLE VII
Suite E5: Full-Scale NF-UNSW-NB15-v2 ($N_{\text{test}} = 298,154$ Flows, Set C) Per-Category Recall, Mondrian Coverage ($\alpha = 0.10$), and Selective SOC Triage Performance ($\tau_{\text{hi}} = 0.25$)

Flow Category	Locked Test Flows (N_{test})	Stage-3 Closed-Set, Stage-5 Conformal, and Stage-6 Selective SOC Triage Metrics				
		Raw Recall (%)	Mondrian Cov (%)	Auto Share ($\mathcal{A}$ %)	Auto Acc ($\mathcal{A}$ %)	Accuracy Lift
Benign	286,750	95.80	89.97	72.06	99.44	+3.64 pp
Exploits	4,486	70.71	89.97	45.27	93.16	+22.45 pp
Fuzzers	3,097	75.94	90.60	61.96	95.52	+19.58 pp
Reconnaissance	1,676	52.03	89.38	37.23	96.63	+44.60 pp
Generic	899	67.41	88.88	47.50	98.36	+30.95 pp
DoS	626	53.04	87.22	24.76	74.84	+21.80 pp
Shellcode	215	76.74	92.09	40.47	98.85	+22.11 pp
Backdoor	125	37.60	90.40	28.00	94.29	+56.69 pp
Analysis	116	33.62	92.24	18.97	100.00	+66.38 pp
Known Mean	—	62.54	90.08	41.80	94.57	+**32.02** pp
Zero-Day (Worms)	164	9.76	9.76	13.41	0.00	86.59% Escalated[a]

[a]Global Telemetry: Automated Coverage ($\mathcal{A}$) = 71.09% (211,954 flows), Automated Accuracy = 99.31%, FAR = 4.20%, AURC = 0.0597.

TABLE VIII
Suite E5: Volumetric Shift Sensitivity Sweep Comparing Unconstrained Versus Monotone-Constrained GBDT

Budget ε	Attack Recall (%)		Macro-F1 Score	
	Unconstr.	Monotone	Unconstr.	Monotone
0.00	99.96	100.00	0.9939	0.9943
0.25	99.87	100.00	0.9934	0.9943
0.50	99.73	100.00	0.9929	0.9943
0.75	98.88	100.00	0.9891	0.9943
1.00	0.59	100.00	0.3677	0.9943

trees, unified-backbone Mahalanobis Weibull EVT open-set recognition, and binary-search accelerated Inductive Mondrian Conformal Prediction, our system eliminates minority-class coverage collapse (+49.14 percentage points on rare Analysis attacks), automates 71.09% of enterprise traffic triage at 99.31% accuracy (AURC = 0.0597), escalates 86.59% of unseen zero-day attacks to human analysts, and provably prevents upper-tail volumetric evasion collapse.

Acknowledgment

The author thanks the University of Queensland Cyber Security Research Group for maintaining the standardized NetFlow v2 benchmark repositories used in this study.

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